⑥ Analyze Big Endian and Little Endian methods of byte addressing with example?

Ans:

In a computer system, data is stored in memory in the form of bytes - when multi-byte data such as integers are stored, the order in which bytes are arranged is called endianness.

Big Endian:-

In Big Endian, the most significant byte is stored at lowest memory address, and the least significant byte is stored at the highest address.

Example:

Consider a 32-bit number 0x12345678

Address	Data
1000	12
1001	34
1002	56
1003	78

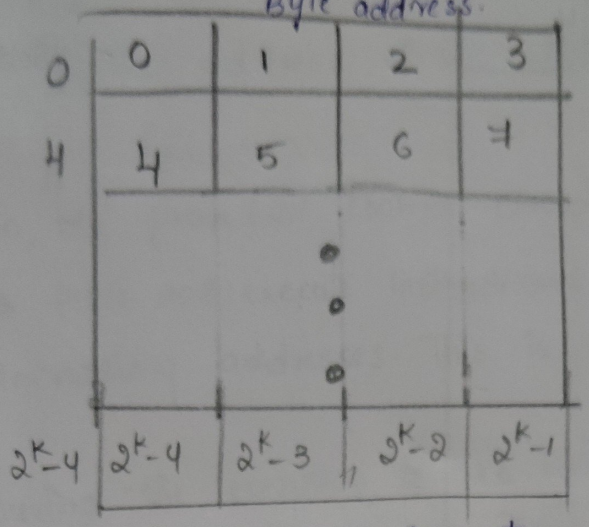
points:

→ MSB stored first

→ Easy for humans to read

→ used in network communication

In case, byte addresses 0, 4, 8 ... are taken as the addresses of successive words in the memory



Big-endian assignment.

Little-endian:-

In Little Endian, the least significant byte is stored at the lowest memory address, and the MSB is stored at the highest address.

Example:-

For the same number 0X 12345678.

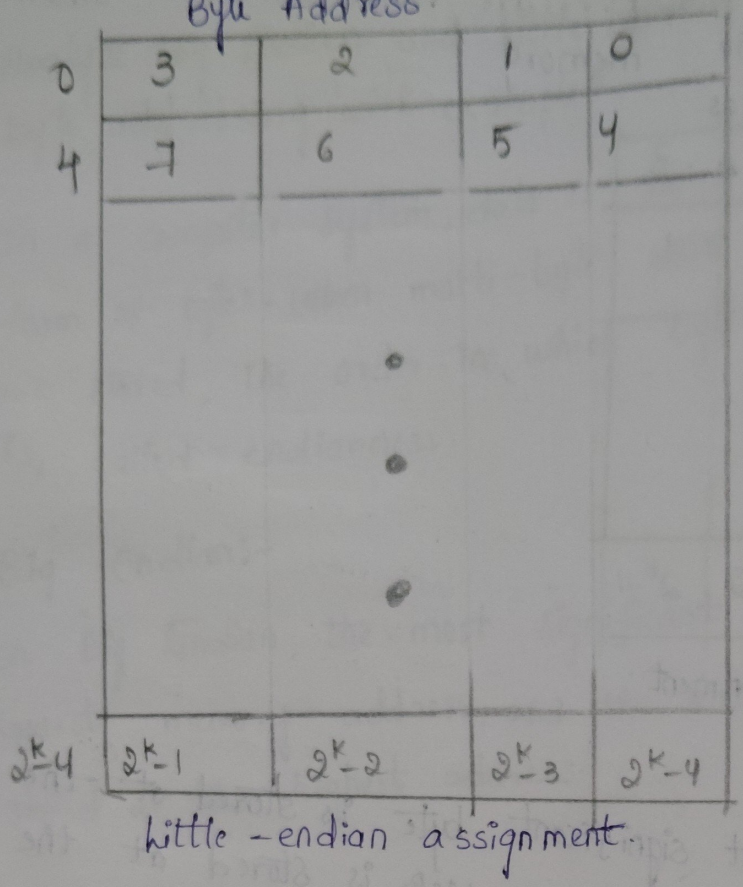
Address	Data
1000	78
1001	56
1002	34
1003	12.

points:-

→ LSB stored first

→ Efficient for arithmetic operations

→ Used in Intel and AMB processors.



Little-endian assignment.

Comparison:-

feature	Big-Endian	Little Endian
Byte stored first	MSB	LSB
Readability	Easy	Difficult
Usage	Networks	personal computers.

7) Discuss Instruction Execution and straight line sequencing?

* The program is executed as follows:

1) Initially, the address of the first instruction is loaded into pc

2) Then, the processor control circuits use the information in the pc to fetch and execute instructions, one at a time, in the order of increasing addresses. This is called straight-line sequencing.

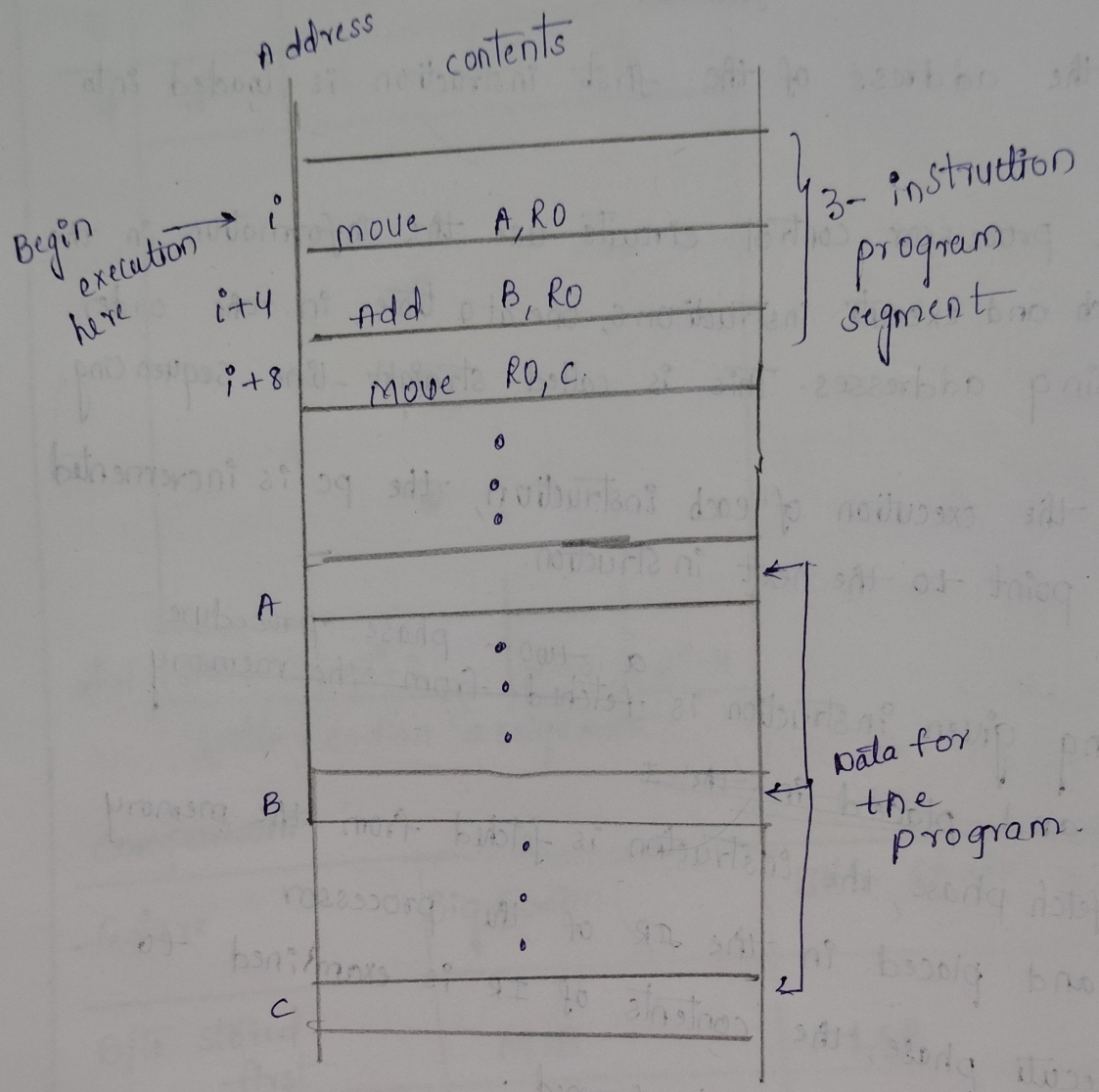
3) During the execution of each instruction, the pc is incremented by 4 to point to the next instruction.

4) executing given instruction is ~~fetched from the memory location and placed in the I~~ a two-phase procedure.

(i) In fetch phase, the instruction is fetched from the memory location and placed in the IR of the processor

(ii) In execute phase, the contents of IR is examined to determine which operation performed.

→ The specified operation is then performed by the processor.



(8) Explain one address, two address, three address with examples. Also use these instructions to carry out C x [A] + [B]

Ans: One-Address Instructions:

A one-address instruction uses only one explicit address. The second operand and the result are implicity stored in a special register called the Accumulator.

Explanation:-

→ one operand is specified by the address field.

→ The other operand is assumed to be in the accumulator.

→ The result is stored back in the accumulator.

Advantages:-

→ Simple instruction format

→ Less memory required for instruction storage.

Disadvantages:-

→ More instruction required

→ Frequent use of accumulator increases execution time.

one-Address instructions to perform C ← [A] + [B]

```
LOAD A  ;  AC ← M[A]
ADD B   ;  AC ← AC + M[B]
STORE C ;  M[C] ← AC.
```

Two-Address instruction:-

Two-address instruction contains two explicit addresses. one operand is also used as the destination.

Explanation:-

→ one address specifies both source and destination.

→ Result replaces one of the operands.

Advantages:-

→ Fewer instructions than one-address

→ Faster execution.

Disadvantages:-
→ one operand is overwritten.
→ slightly complex instruction format

Two-Address Instructions - to perform C ← [A] + [B].

```
MOV A, C   ; C ← A
ADD B, C   ; C ← C+B.
```

Three-Address Instruction:-
A three-address instruction uses three explicit addresses: two source operands and one destination.

Explanation:-
→ Two operands are read from different locations.
→ Result is stored in a separate destination
→ No operand is destroyed.

advantages:-
→ fewer instructions
→ fast execution
→ clear and flexible.

Disadvantages:-
→ Larger instruction size
→ Requires more memory.

Three-Address Instruction to perform C ← [A] + [B]

```
ADD C, A, B.
```

Explain different types of instruction formats with examples

→ 8th question answer only for this question also.

An instruction format refers to the layout of bits in an instruction that tells the computer.

* opcode → specifies the operation to be performed

* operand → specify data or addresses.

* Addressing mode →

Based on the no. of addresses present in an instruction, instruction formats are classified into different types.

Describe the concept of Branching with an example program of instruction execution?

→ The loop is a straight line sequence of instructions executed as many times as needed. It starts at location LOOP and ends at the instruction Branch >0.

→ During each pass through this loop, the address of the next list entry is determined, and that entry is fetched and added to R0.

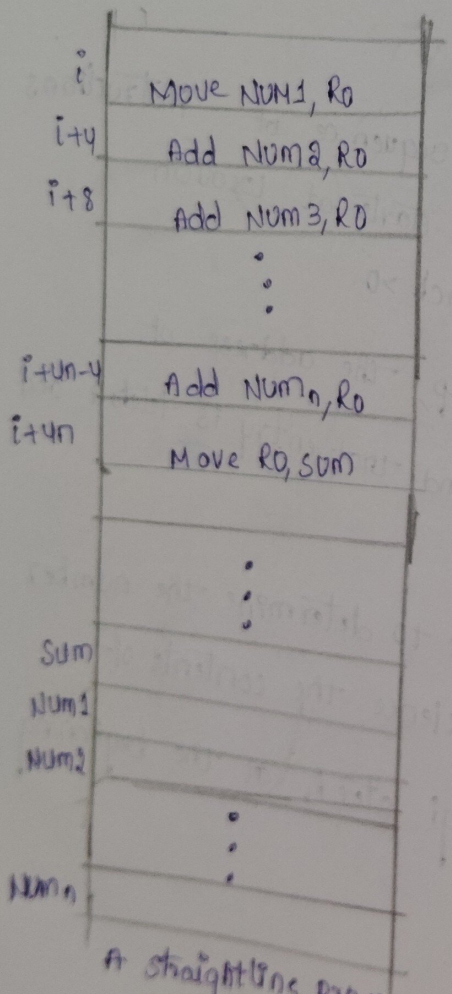
→ Register R_1 is used as a counter to determine the number of times the loop is executed. Hence, the contents of location N are loaded into register R_1 at the beginning of the program.

→ within the body of the loop, the instruction Decrement R_1, reduces the contents of R_1 by 1 each time through the loop

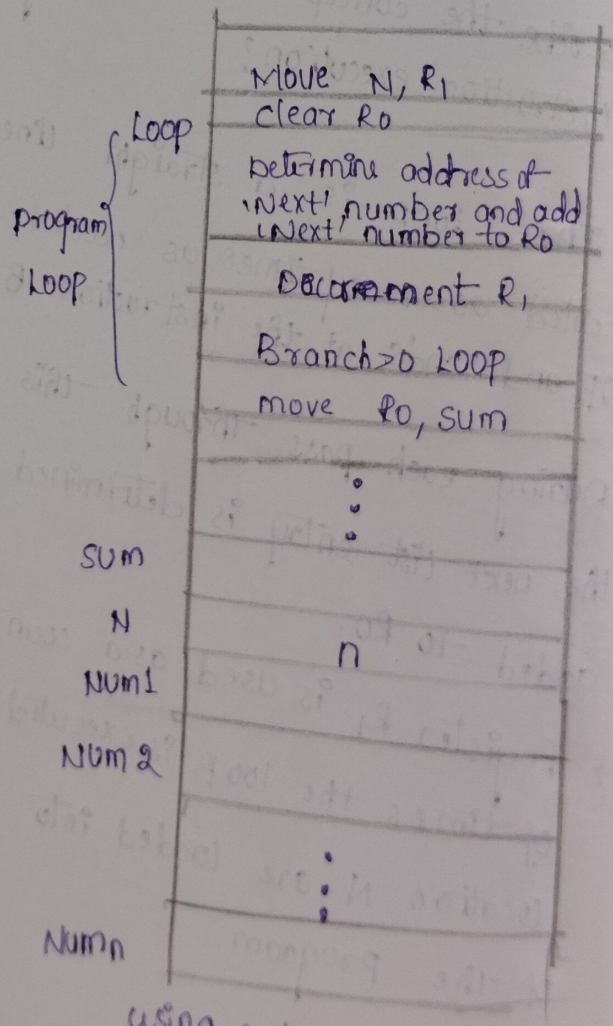
→ The Branch instruction loads a new value into the program counter. As a result, the processor fetches and executes the instruction at this new address called the branch target.

→ A conditional branch instruction causes a branch only if a specified condition is satisfied.

→ If the condition is not satisfied, the pc is incremented in the normal way, and the next instruction in sequential address order is fetched and executed.



A straightline program for adding n numbers



using a loop to add n numbers.